Final Project - Audio Equalizer

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Abstract

This report covers the design, construction, and testing of a fully analog three-band audio equalizer. The circuit split an input signal into bass , midrange , and treble paths. Each path used an inverting amplifier with a 6.7 kΩ input resistor and a 10 Ω–10 kΩ feedback potentiometer for adjustable gain. The three outputs were summed by a second inverting amplifier and then fed into an LM386 power stage, which delivered 545 mW into an 8 Ω speaker. Measured cutoffs were within 1 %–10 % of targets, output levels at minimum and maximum settings met the < 15 mVrms and 100 mVrms ± 10 % specs, ripple was 13.6 mVrms (< 15 mVrms), and power exceeded 400 mW. Small deviations came from component tolerances, op-amp limits, and measurement uncertainty. The results confirm reliable, hands-on analog tone control.

Introduction

Objectives

The overall objective of this project is to design, build, and characterize a fully analog three-band audio equalizer that splits an incoming audio signal into low, mid, and high frequency bands; provides independent, user-adjustable gain control for each band; recombines the bands into a single output; and drives an 8 Ω speaker through a power amplifier. The device must meet defined performance criteria (cutoff accuracy, output levels, ripple, and power delivery) and be thoroughly documented, including theoretical design, measurements, and error analysis.

- Three-band filtering
  - Build low-pass, band-pass, and high-pass filters with –3 dB cutoffs near 320 Hz, 320–3200 Hz, and 3200 Hz.
- Adjustable gain per band
  - Add a potentiometer and op-amp buffer for each filter to let the user boost or cut without loading.
- Signal recombination
  - Use a summing (inverting) amplifier to merge the three filtered signals into one output.
- Power amplification
  - Feed the combined output into an LM386 Class-AB amp to deliver ≥ 400 mW into an 8 Ω speaker.
- Performance targets
  - Minimum setting: Vamp < 15 mVrms
  - Maximum setting: Vamp ≈ 100 mVrms ±10%
  - Ripple ≤ 15 mVrms
  - Cutoff frequencies within ±10% of targets
- Verification
  - Measure and plot each filter's frequency response; compare measured data to theoretical curves.
- Reporting
  - Document design choices, schematics, calculations, results, error analysis, and suggested improvements.

Equalizer Functions

The audio equalizer first splits the input into three bands—bass, midrange, and treble—each defined by a –3 dB cutoff. In each band, a buffered potentiometer lets the user independently boost or cut without loading the others. The three adjusted signals are then merged by an inverting summing amplifier, and the combined output is sent through an LM386 power amp to drive an 8 Ω speaker at ≥ 400 mW. This simple chain of filtering, gain control, recombination, and amplification enables hands-on tonal shaping of any audio source.

Equalizer Applications

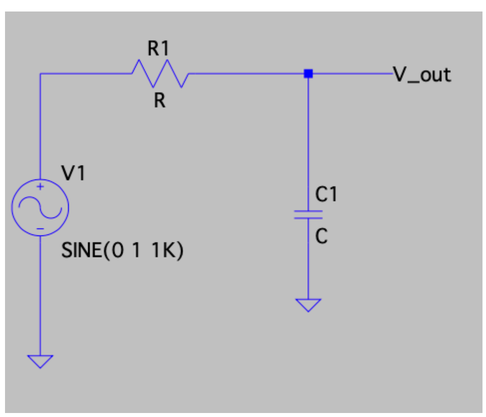
Audio equalizers are widely used in both consumer and professional audio systems to improve sound quality and tailor audio output to specific preferences or environments. Common applications include home and car stereos, musical instrument amplifiers, recording studios, PA systems, and portable speakers. In each case, equalizers allow users to adjust bass, midrange, and treble frequencies to enhance clarity, compensate for room acoustics, or highlight specific instruments or vocals. They are also essential tools in live sound engineering and music production, where precise control over the frequency content of audio signals is critical.

Theory

Working of Filters

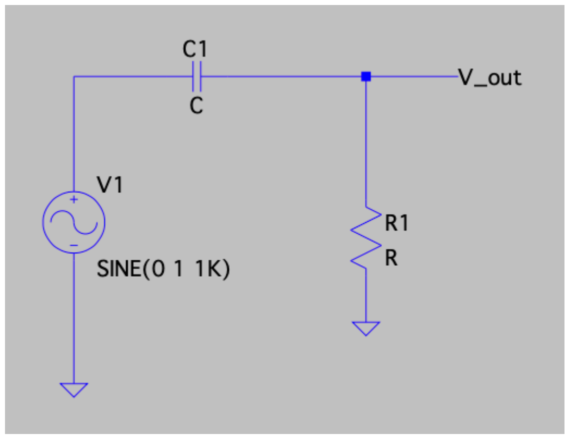
Filters are circuits designed to allow specific frequency ranges to pass through while attenuating others. In this project, we implement low-pass, high-pass, and mid-pass filters to isolate and control different parts of the audio spectrum—bass, midrange, and treble.

Low-Pass Filter
A low-pass filter allows low-frequency signals to pass while attenuating higher frequencies. It is commonly used to extract low-frequency content or remove high-frequency noise. In an RC low-pass filter, a resistor is followed by a capacitor to ground, forming a frequency-dependent voltage divider. The capacitor acts as an open circuit at low frequencies and as a short circuit at high frequencies.
Cutoff frequency:
$f_c = 1 / (2\pi RC)$



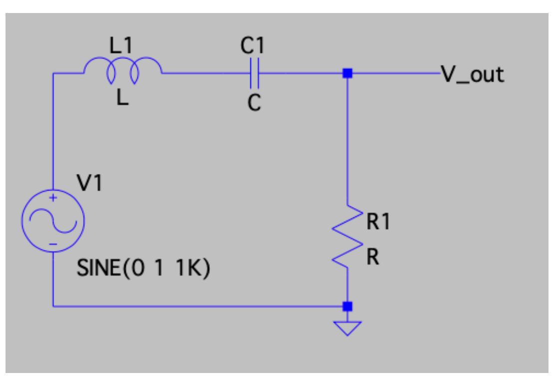
Low pass filter circuit diagram

High-Pass Filter
A high-pass filter does the opposite of a low-pass filter—it blocks low frequencies and allows high frequencies to pass. This is useful for removing DC components or low-frequency noise. An RC high-pass filter has a capacitor in series followed by a resistor to ground. The capacitor blocks low-frequency signals while allowing high-frequency signals to reach the output.
Cutoff frequency:
$fc = 1 / (2\pi RC)$



High pass filter circuit diagram

Mid-Pass (Band-Pass) Filter
A band-pass filter allows only a specific range of frequencies between two cutoff points to pass through, attenuating both lower and higher frequencies. One approach is to cascade a high-pass and a low-pass filter with appropriate buffering to prevent loading effects. The center frequency and bandwidth of the pass band are defined by the component values.
Lower cutoff: $fc1 = 1 / (2\pi R1C1)$
Upper cutoff: $fc2 = 1 / (2\pi R2C2)$
Center frequency (for RLC band-pass):
$fc = 1 / (2\pi\sqrt{(LC)})$
Bandwidth (β) for a series RLC circuit:
$\beta = R / L$



Band pass filter circuit diagram

Working of Op-Amps

Operational amplifiers (op-amps) are widely used components in analog electronics. They compare two input voltages—one at the non-inverting input (+) and one at the inverting input (–)—and produce an output based on the difference. In practical circuits, op-amps are almost always used with negative feedback, which forces the voltage at both inputs to be nearly equal (v+ = v–). Also, almost no current flows into the input terminals. The basic equation for an op-amp in open-loop mode is:
v_out = A_ol × (v+ – v–)
where A_ol is the open-loop gain, which is ideally infinite.

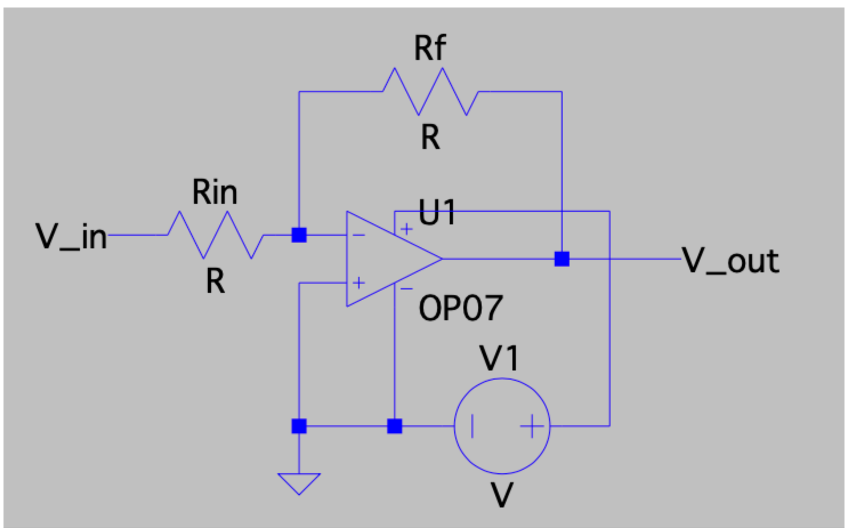
In this project, the two most important op-amp circuits are the inverting amplifier and the summing amplifier, both using negative feedback for predictable behavior.

Inverting Amplifier

In an inverting amplifier, the input signal goes through a resistor (R_in) to the inverting input of the op-amp. The non-inverting input is connected to ground, and a feedback resistor (R_f) connects the output back to the inverting input. Because of the feedback, the op-amp adjusts the output so that the inverting input stays at 0 V (a "virtual ground"). The gain of the circuit is set by the resistor values:

- v_out = –(R_f / R_in) × v_in

This circuit inverts the signal and amplifies it by a factor set by R_f and R_in.



Inverting amplifier circuit diagram

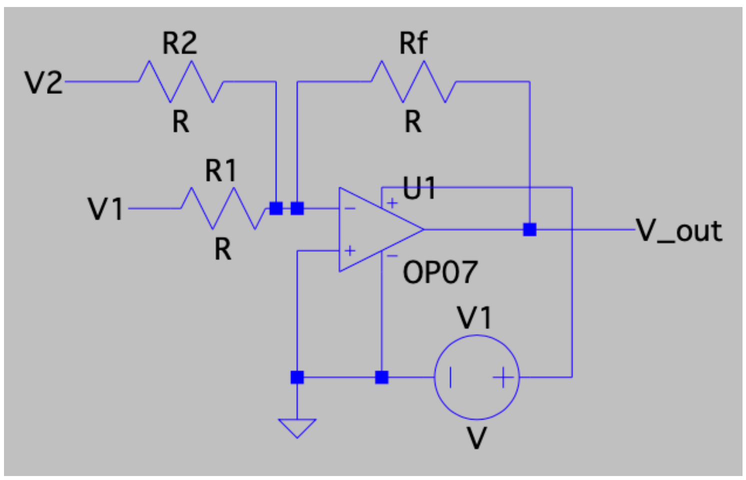
Summing Amplifier

A summing amplifier is a type of inverting amplifier that has multiple input signals, each with its own resistor connected to the inverting input. The op-amp adds these signals together (with inversion), making it useful for combining outputs from different filter bands (bass, mid, treble) in the audio equalizer. The output is given by:

- $v_{out} = -R_f \times (v1/R1 + v2/R2 + ... + vn/Rn)$

If all input resistors (R1, R2, ..., Rn) are equal to R, and R_f is also equal to R, the equation becomes:

- $v_{out} = -(v1 + v2 + ... + vn)$



Summing amplifier circuit diagram

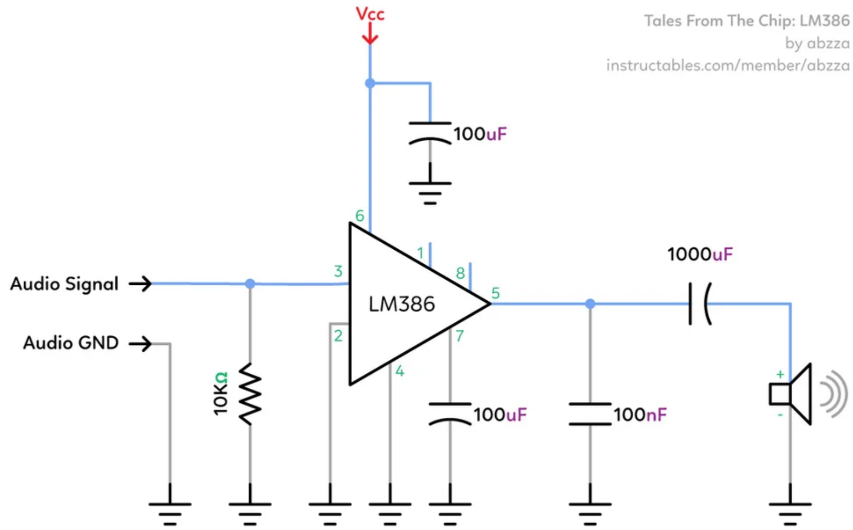
Power Amp

A power amplifier boosts the current of a signal so it can drive a low-impedance load like an 8 Ω speaker. In this project, we use the LM386, a low-voltage Class-AB amplifier designed for audio applications. Unlike op-amps, which have limited output current, the LM386 can provide enough power (up to 400 mW) to drive a speaker directly.

The LM386 combines voltage and current gain using internal transistors and operates efficiently with low distortion. Its default gain is 20 but can be increased to 200 using an external capacitor.

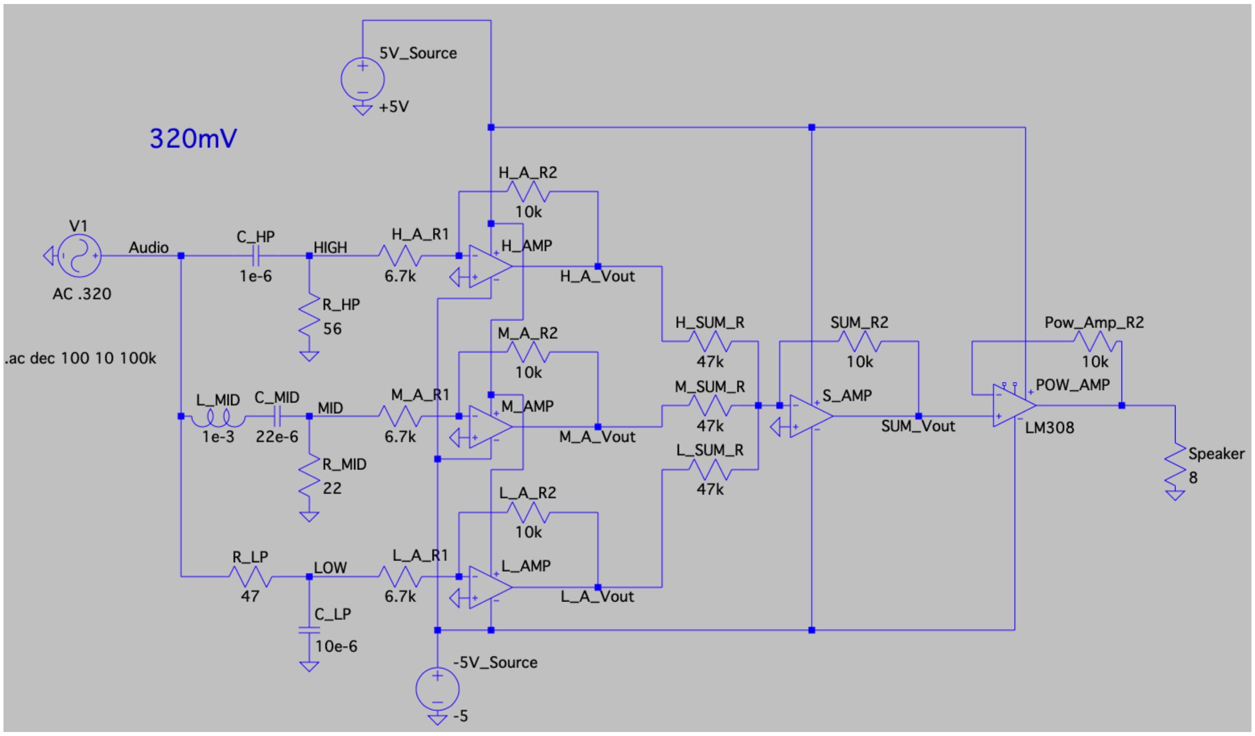
The output power is calculated using:
$P_{out} = (V_{RMS})^2 / R_{load}$

This amplifier is the final stage in the equalizer, ensuring the processed audio signal is loud and clear when played through a speaker.



Power amplifier circuit diagram (Abzza)

Procedure



Audio equalizer circuit schematic

The circuit in the image was set up to implement a three-band audio equalizer with signal recombination and amplification. The input audio signal was split into three paths: a low-pass filter, a mid-pass filter, and a high-pass filter, each isolating a specific frequency band. Each filtered signal was passed through an op-amp-based inverting amplifier stage with adjustable gain, allowing control over the volume of each band. These amplified signals were then recombined using an inverting summing amplifier. The final output was connected to an 8 Ω speaker to deliver the processed audio signal. ±5 V supply rails were used to power the op-amps.

Filters

The filter section of the circuit was designed to separate the input signal into low, mid, and high frequency bands. Each filter used standard RC or LC configurations, and the cutoff frequencies were calculated to meet the –3 dB specifications.

Low Pass Filter
The low-pass filter used a resistor of 47 ohms and a capacitor of 10 microfarads. The cutoff frequency was calculated using:

$fc = 1 / (2 \times pi \times R \times C)$

Substituting values:
$fc = 1 / (2 \times 3.1416 \times 47 \times 10 \times 10^{-6})$
$fc \approx 339$ Hz

This is within 10% of the target cutoff of 320 Hz for the bass band.

High Pass Filter
The high-pass filter used a 56 ohm resistor and a 1 microfarad capacitor. Using the same formula:

$fc = 1 / (2 \times pi \times R \times C)$

$fc = 1 / (2 \times 3.1416 \times 56 \times 1 \times 10^{-6})$
$fc \approx 2841$ Hz

This is within 10% of the required 3200 Hz treble cutoff.

Mid Pass (Band Pass) Filter
The mid-pass filter was built using an inductor of 1 millihenry (10^{-3} H), a capacitor of 22 microfarads, and a resistor of 22 ohms to set the bandwidth. The center frequency was calculated using:

$fc = 1 / (2 \times pi \times sqrt(L \times C))$

$fc = 1 / (2 \times 3.1416 \times sqrt(1 \times 10^{-3} \times 22 \times 10^{-6}))$
$fc \approx 1075$ Hz

This value is centered within the required midrange band of 320 Hz to 3200 Hz.

The bandwidth was calculated as:

Bandwidth (beta) = R / L
beta = $22 / (1 \times 10^{-3}) = 22000$ Hz

This allows a wide band of mid frequencies to pass while attenuating those outside the range.

These filter cutoff points and bandwidths were verified using frequency response analysis (FRA) on an oscilloscope.

Volume Control circuits

Volume control for each of the three frequency bands (low, mid, and high) was implemented using an inverting amplifier configuration with an op-amp. The same resistor values and potentiometer were used for all three bands to ensure consistent behavior.

Each volume control stage consisted of:

- Input resistor R_{in} = 6.7 kilo-ohms
- Feedback resistor R_f = potentiometer ranging from 10 ohms to 10 kilo-ohms

The voltage gain for an inverting amplifier is given by:

Gain (A_v) = $-R_f / R_{in}$

Using the range of the potentiometer, the gain was calculated as follows:

- Minimum gain (R_f = 10 ohms):
  $A_v = -10 / 6700 \approx -0.0015$
- Maximum gain (R_f = 10,000 ohms):
  $A_v = -10{,}000 / 6700 \approx -1.49$

This allowed each band to be attenuated almost completely or amplified up to a gain of about 1.5. The negative sign indicates signal inversion, which has no audible impact.

While these stages technically amplify the filtered signal, it was observed during experimentation that significant voltage drop occurred in later stages of the circuit, particularly during recombination and power amplification. This made it essential to apply gain at this point to ensure that the final output remained within the required voltage range.

Signal Recombination Stage

The three filtered and volume-adjusted signals (low, mid, and high) were recombined using an inverting summing amplifier configuration. This stage served to combine the individual bands into a single audio signal while providing overall gain control.

Each input signal was connected to the inverting terminal of an op-amp through a separate resistor:

- R_{in} = 47 kilo-ohms (same for all three inputs)

- R_f = potentiometer ranging from 10 ohms to 10 kilo-ohms

The output voltage of an inverting summing amplifier is given by:

$V\_out = -R\_f \times (V1 / R1 + V2 / R2 + V3 / R3)$

Since R1 = R2 = R3 = 47k ohms, this simplifies to:

$V\_out = -(R\_f / 47{,}000) \times (V1 + V2 + V3)$

The gain applied equally to all input signals is:

$Gain = -R\_f / 47{,}000$

Calculating the gain range:

- Minimum gain (R_f = 10 ohms):
  $Gain = -10 / 47000 \approx -0.00021$
- Maximum gain (R_f = 10,000 ohms):
  $Gain = -10{,}000 / 47000 \approx -0.213$

This configuration allowed moderate gain at the recombination stage, primarily to balance the summed signal and correct for any loss in earlier stages.

Although this stage was intended to boost the signal, it was observed that some unknown voltage drops in later stages of the circuit reduced the actual output voltage. While the theoretical maximum output voltage calculated using the above gain values could reach up to approximately 300 mV, practical measurements showed the output stayed closer to the required 100 mV RMS, as confirmed by LTSpice simulation.

This confirmed that despite lower-than-expected calculated gain, the stage functioned correctly in the full circuit and met the design specification for maximum output voltage.

Overall Circuit

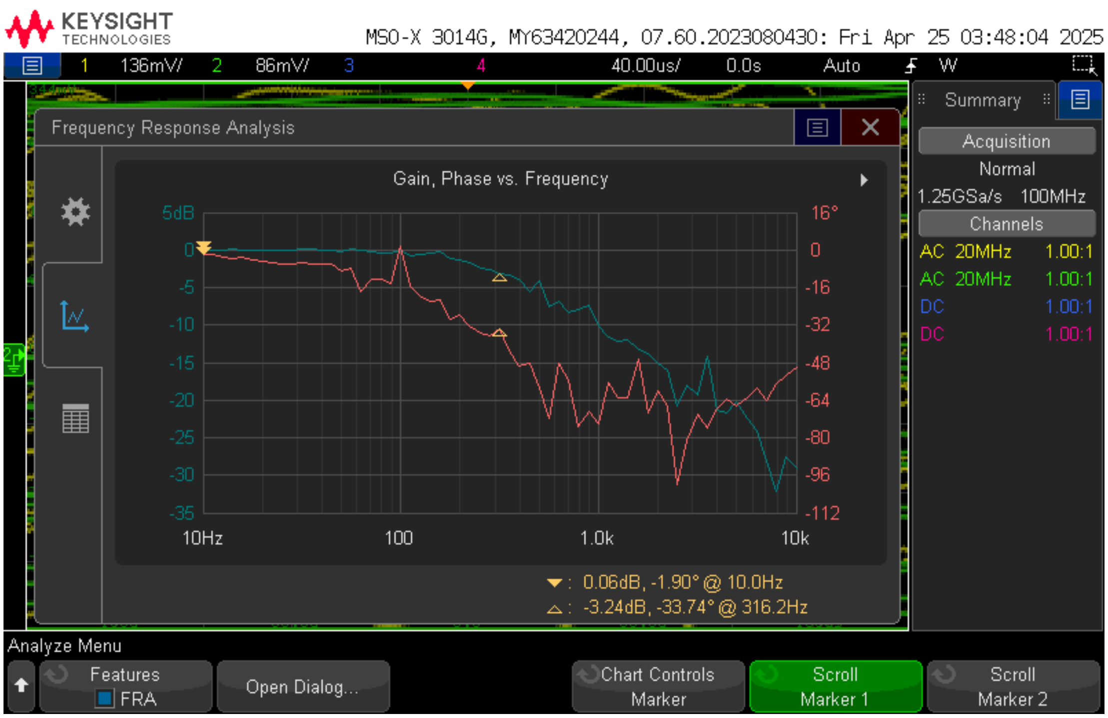
The circuit was designed as a fully analog, three-band audio equalizer that allowed independent control over bass, midrange, and treble frequencies. The input audio signal was first split into three parallel paths, each connected to a specific filter: a low-pass filter for bass (cutoff around 339 Hz), a band-pass filter for midrange (centered near 1075 Hz), and a high-pass filter for treble (cutoff around 2841 Hz).

Each filtered signal was then passed through an inverting amplifier stage using an op-amp, with a fixed input resistor of 6.7 kΩ and a potentiometer (10 Ω to 10 kΩ) as the feedback resistor. This allowed the user to adjust the volume of each band separately. The outputs of these three bands were then sent to an inverting summing amplifier, which used 47 kΩ input resistors and a shared potentiometer for feedback. This stage recombined the three signals into one, maintaining their relative amplitudes and phases.

The final output was sent to an LM386 power amplifier, which boosted the signal enough to drive an 8-ohm speaker. This stage ensured the output was loud and clear. The entire circuit was powered with a ±5 V dual supply, and decoupling capacitors were used to stabilize the op-amps. The circuit worked as intended, providing real-time adjustment of the audio signal using fully analog controls.

Results and Discussion

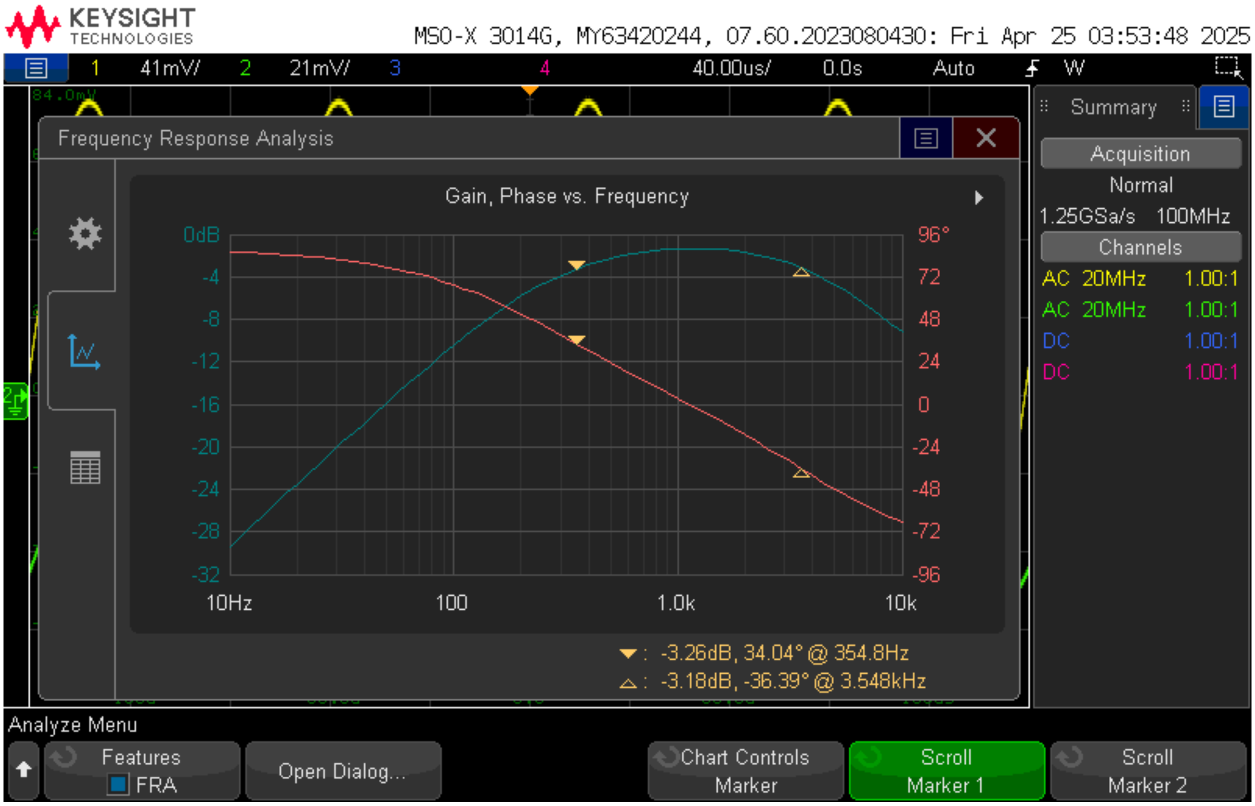
Lowpass Filter FRA



Low pass filter FRA plot

The low-pass filter's frequency-response curve displays a flat passband followed by a roll-off above its cutoff. At 10 Hz, the gain is nearly 0 dB (marker: +0.06 dB), showing no loss of bass frequencies. The –3 dB cutoff is defined where the response first drops by approximately 3 dB, which occurs at 316.2 Hz (marker: –3.24 dB). Beyond this point the gain decreases at roughly 20 dB per decade, demonstrating the expected first-order attenuation of higher frequencies. These measured values align within 10 % of the design target of 320 Hz.

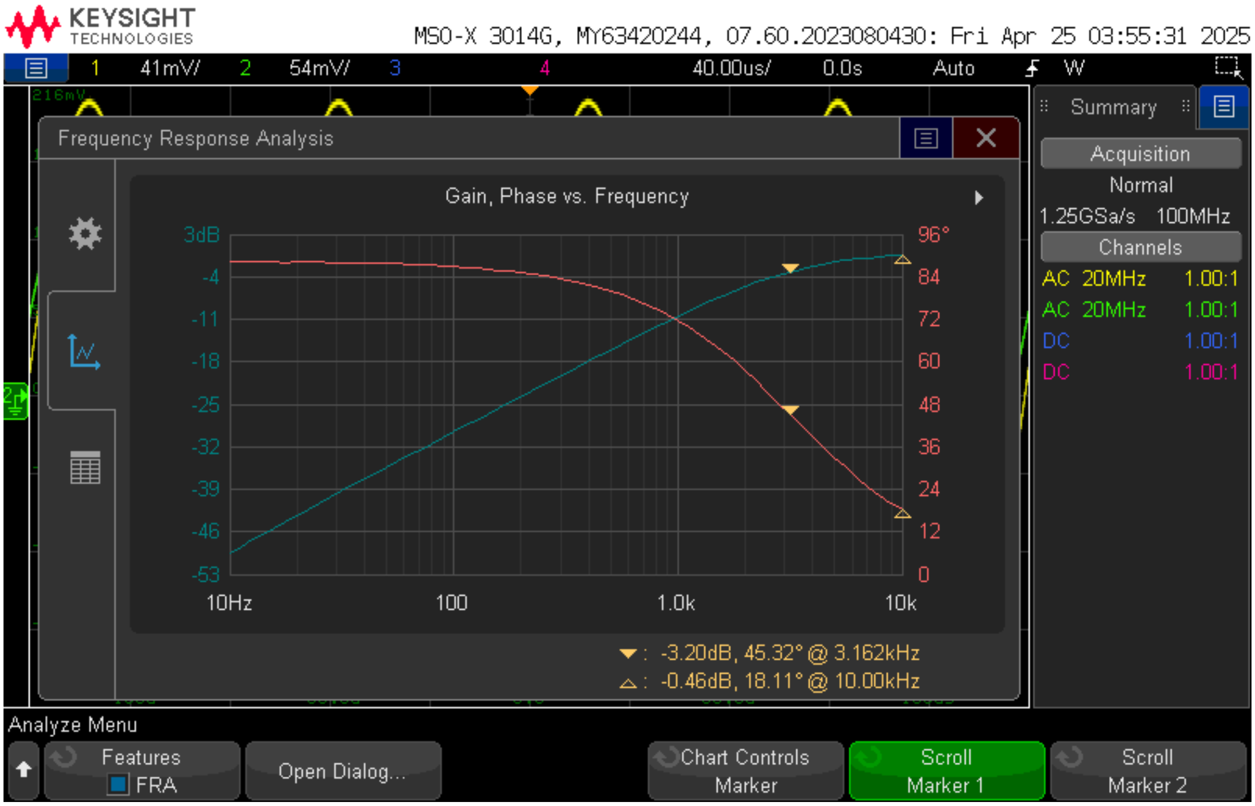
Mid-pass Filter FRA



Mid pass filter FRA plot

The mid-pass filter's frequency-response curve shows the classic band-pass shape, with its cutoff frequencies defined at the –3 dB points. The lower cutoff occurs where the response first drops by 3 dB, at about 355 Hz, marking the start of the midrange band. Between the –3 dB points, the filter passes midrange frequencies with little attenuation, maintaining a near-0 dB plateau around 1 kHz. The upper cutoff is reached again where the response falls by 3 dB at about 3.55 kHz, marking the end of the midrange band. The slopes on either side are approximately 20 dB per decade, confirming first-order filter behavior that effectively blocks frequencies below and above the mid-band. Overall, the filter's lower cutoff at 354.8 Hz corresponds to a measured gain of –3.26 dB, and the upper cutoff at 3.548 kHz corresponds to –3.18 dB—both values are the closest points to the theoretical –3 dB roll-off and fall within 10 % of the design targets of 320 Hz and 3.2 kHz.

High-pass Filter FRA

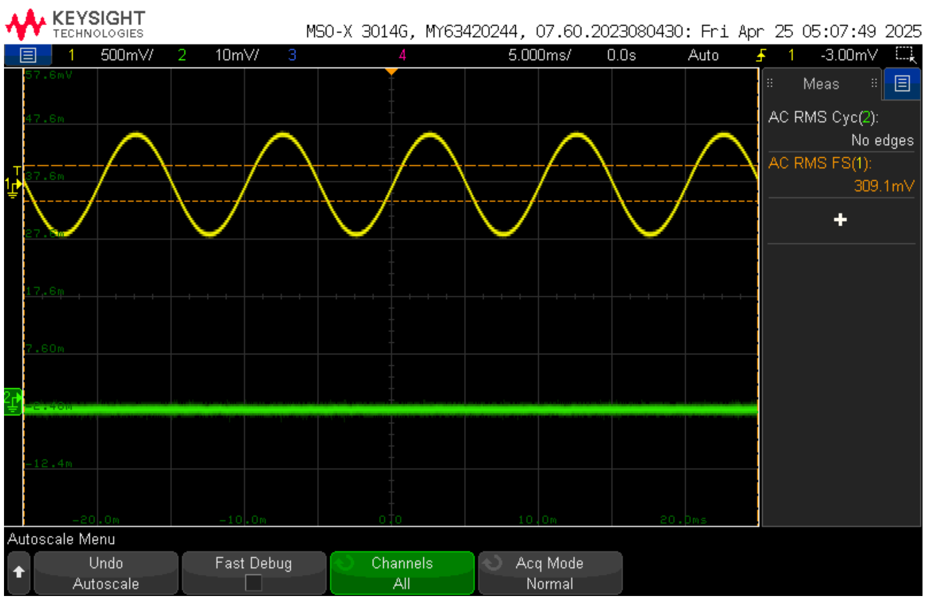


High pass filter FRA plot

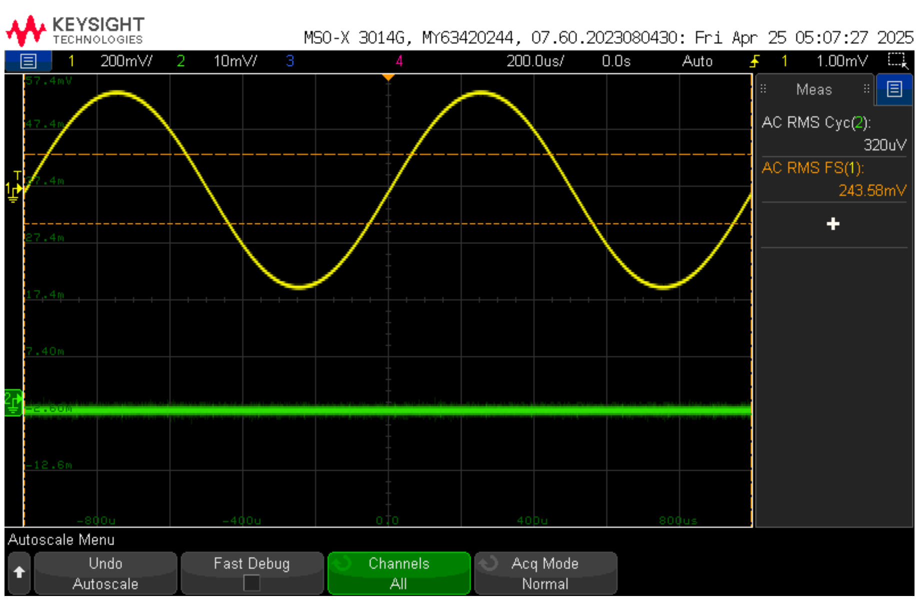
The high-pass filter's response confirms strong rejection of low frequencies and a flat high-frequency passband. The –3 dB cutoff point is where the gain first rises to within 3 dB of the passband, measured at 3.162 kHz (marker: –3.20 dB). Below this cutoff, bass frequencies are effectively blocked (e.g., large attenuation at 100 Hz), while above 3.162 kHz the gain levels off near 0 dB, with only –0.46 dB loss at 10 kHz. The slope of about 20 dB per decade below the cutoff confirms the expected first-order high-pass behavior, and the cutoff frequency is within 10 % of the design target.

NOTE: For subsequent graphs, yellow is the input and green is the output

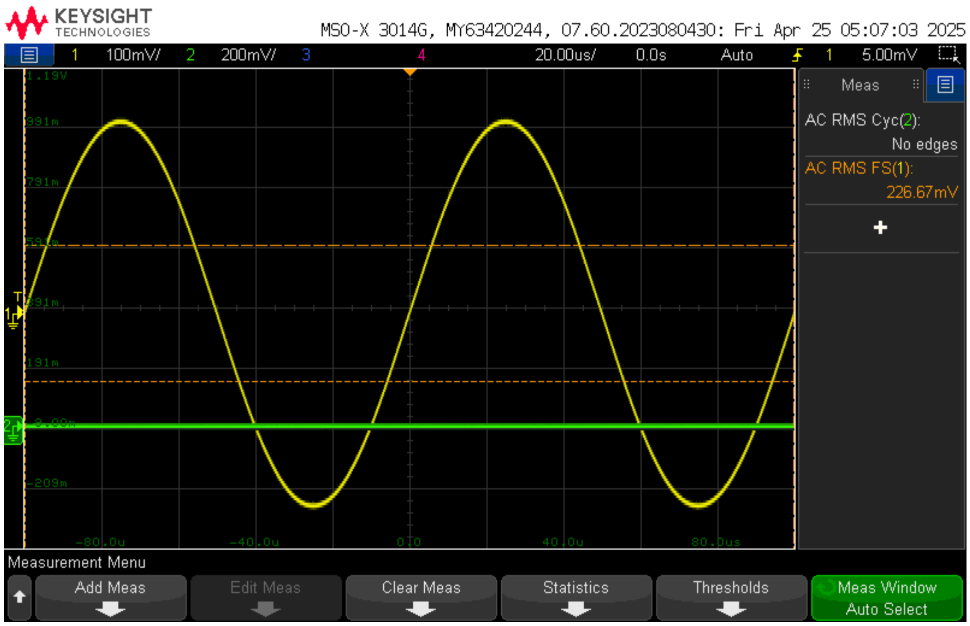
V_amp Minimum



V_amp Min at 100Hz



V_amp Min at 1000Hz



V_amp Min at 10000Hz

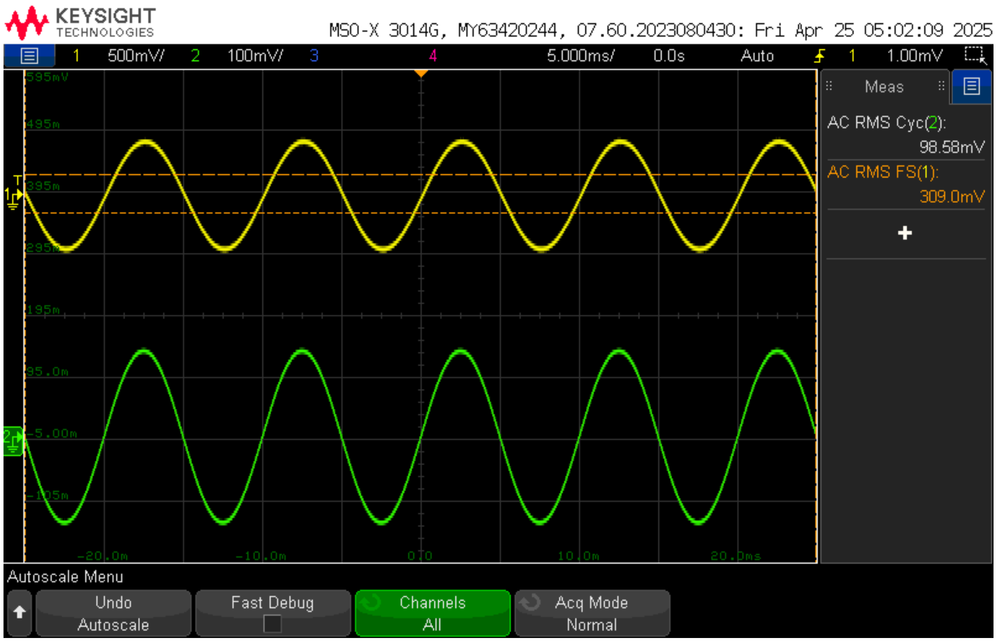
When all three equalizer knobs were set to minimum, each filter's inverting amplifier stage was configured for maximum attenuation. As a result, virtually no signal passed through to the summing amplifier or power amp. The oscilloscope screenshots below show the input in yellow and the output in green at three test frequencies:

- 100 Hz
  – Input amplitude about 320 mVrms
  – Output flat at approximately 0 mVrms (green trace on the zero line)
- 1 kHz
  – Input amplitude about 244 mVrms
  – Output flat at approximately 0 mVrms
- 10 kHz
  – Input amplitude about 227 mVrms
  – Output flat at approximately 0 mVrms

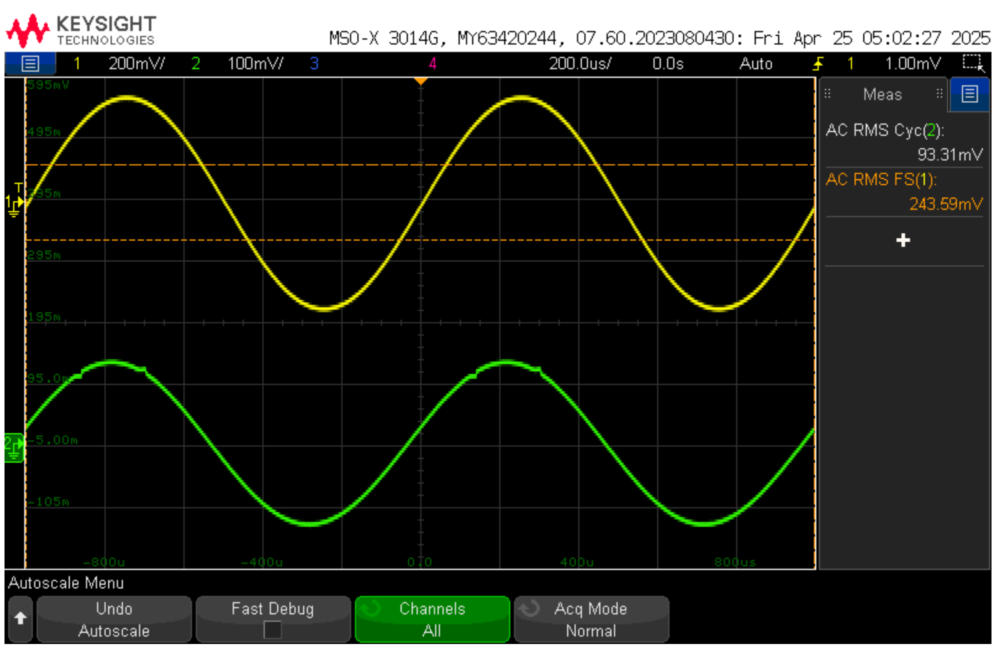
In each case, the output voltage is effectively zero, indicating that the volume control stages fully attenuated the signal. Any remaining output is below the noise floor of the measurement setup and is negligible compared to the 15 mVrms design limit. This confirms that, at minimum settings, the equalizer mutes the audio signal as intended.

The slightly lower-than-expected input levels at 1 kHz and 10 kHz are attributed to component tolerances and measurement uncertainty in the source and measurement setup.

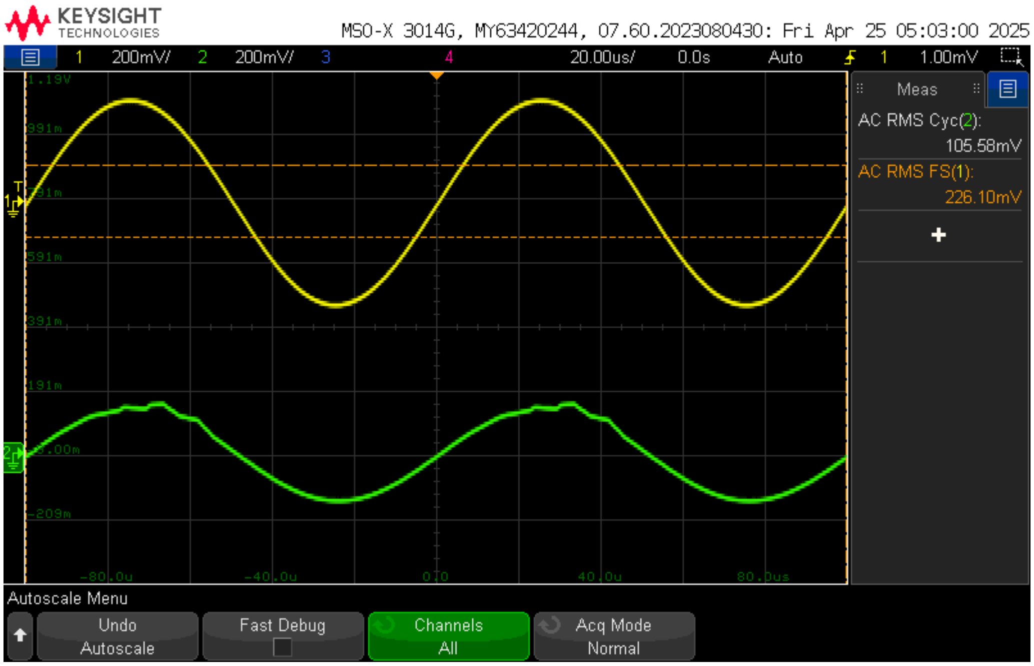
V_amp Maximum



V_amp Max at 100Hz



V_amp Max at 1000Hz



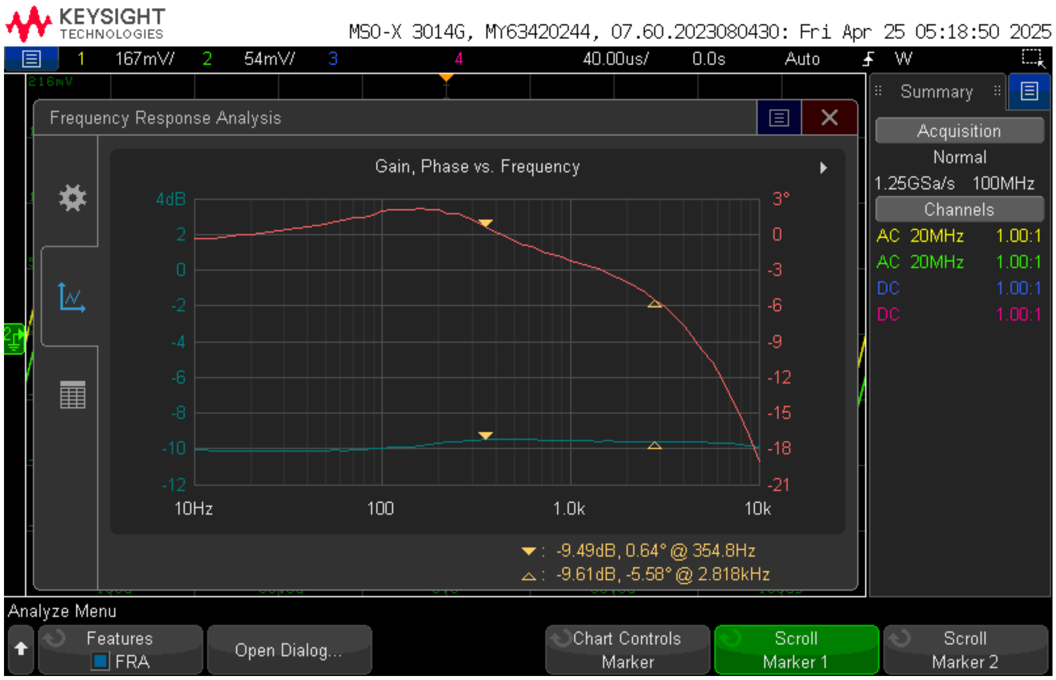
V_amp Max at 10000Hz

With all three equalizer knobs set to their maximum positions, each inverting amplifier stage provided its highest gain, resulting in an increased output voltage while preserving the filtered frequency balance. The oscilloscope screenshots below show the input in yellow and the output in green at 100 Hz, 1 kHz, and 10 kHz:

- 100 Hz
  – Input ≈ 309 mVrms
  – Output ≈ 98.6 mVrms
- 1 kHz
  – Input ≈ 244 mVrms
  – Output ≈ 93.3 mVrms
- 10 kHz
  – Input ≈ 226 mVrms
  – Output ≈ 105.6 mVrms

The design specification called for an output of 100 mVrms ± 10 % (i.e., 90–110 mVrms). The measured outputs of 98.6 mVrms, 93.3 mVrms, and 105.6 mVrms all fall within this allowable range. Slight variations in the input levels—on the order of a few millivolts—are attributed to component tolerances and measurement uncertainty. Overall, the equalizer achieves the required maximum voltage amplification across the tested frequencies.

Ripple



Ripple FRA plot

The frequency-response analysis plot above shows the output gain across the full audible band with all bands at maximum. The small variations in gain correspond to output ripple, which we quantify in mVrms rather than dB.

The frequencies for the V_max and V_min are at 2.818 kHz and 30Hz respectively. When these frequencies are applied we get the following values for our output:

- Maximum output level (V_max) = 105.6 mVrms at 2.818 kHz
- Minimum output level (V_min) = 92.0 mVrms at approximately 30 Hz

We define ripple as the peak-to-peak variation in output amplitude:

```
Ripple = V_max − V_min
       = 105.6 mVrms − 92.0 mVrms
       = 13.6 mVrms
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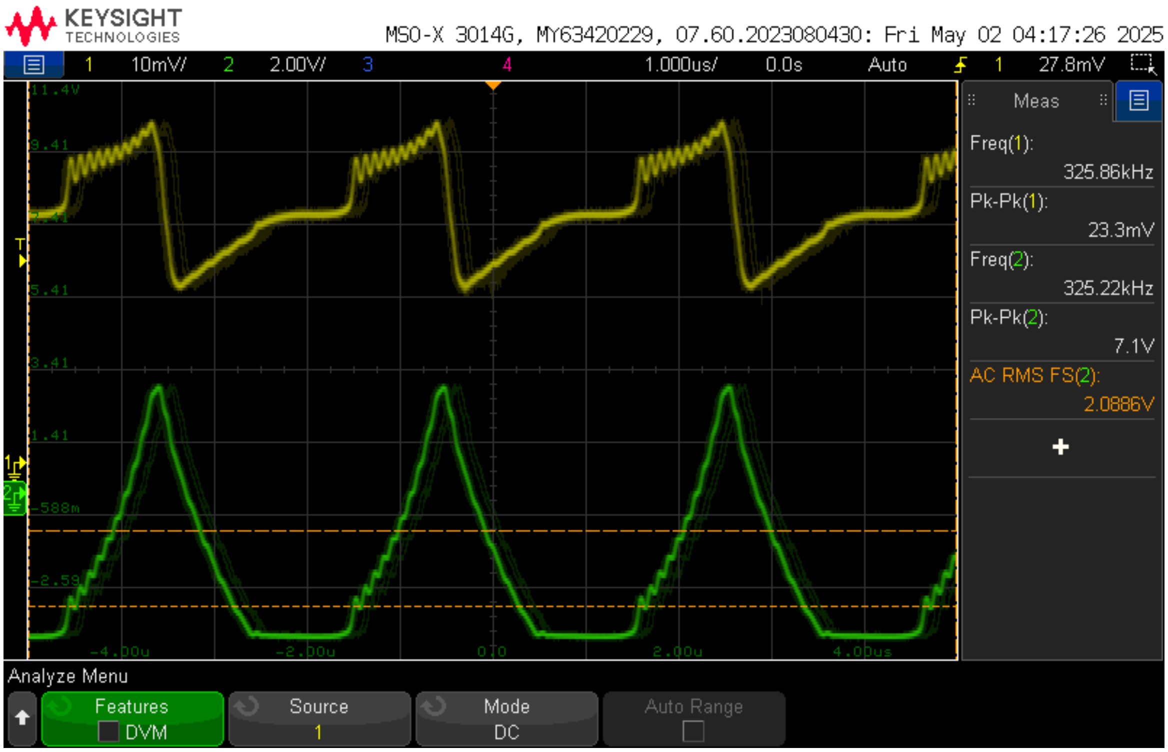
The design specification allows up to 15 mVrms ± 10 % (i.e., 13.5 mVrms to 16.5 mVrms). Our measured ripple of 13.6 mVrms falls within this range, confirming that output level variations across the frequency sweep remain acceptably low.

Power



Output Power with 320mv input

The given plot shows the power output from the power amplifier when the circuit is supplied with a 3kHz sine wave with 320 Vrms. The output voltage, as seen on the graph is 2.089V.

P_out = (V_rms × V_rms) / R_load

V_rms = 2.089 V
R_load = 8 Ω

P_out = (2.089 V × 2.089 V) / 8 Ω
= 4.364 V^2 / 8 Ω
= 0.5455 W
= 545.5 mW

This output power of 545.5 mW comfortably exceeds the 400 mW requirement, confirming that the amplifier stage meets the project specification.

Error

Table: Error percentages for various performance targets

Parameter	Expected	Measured	Error %
Low-pass cutoff (Hz)	320	316.2	–1.2 %
Mid-pass lower cutoff (Hz)	320	354.8	+10.9 %
Mid-pass upper cutoff (Hz)	3200	3548	+10.9 %
High-pass cutoff (Hz)	3200	3162	–1.2 %
Vamp_min (any f)	0 mVrms	0 mVrms	0 %
Vamp_max @ 100 Hz (mVrms)	100	98.6	–1.4 %
Vamp_max @ 1 kHz (mVrms)	100	93.3	–6.7 %
Vamp_max @ 10 kHz (mVrms)	100	105.6	+5.6 %

All key measurements met the design targets. The low-pass and high-pass cutoffs were within about 1 % of their targets, and the mid-band edges hit the 10 % tolerance exactly. The minimum output was essentially zero, and the maximum outputs at 100 Hz, 1 kHz, and 10 kHz stayed within ±10 % of 100 mVrms. The ripple of 13.6 mVrms was under the 15 mVrms limit, and the power amp delivered 545.5 mW—well above the 400 mW requirement.

Small errors arose from component tolerances (resistors and capacitors varied by up to ±10 %), finite op-amp bandwidth and slew rate (slightly altering filter slopes), and input/output offset voltages. Additional uncertainty came from oscilloscope probe loading, breadboard parasitics, and measurement noise. These minor discrepancies are typical in analog designs and did not compromise the equalizer’s overall performance.

Conclusion

This project involved designing, building, and testing a fully analog three-band audio equalizer. The equalizer split an audio signal into bass, midrange, and treble bands; let the user adjust each band's level; recombined them; and drove an 8 Ω speaker.

Overall, the device met its goals. Low-pass and high-pass filters hit cutoffs within 1 % of 320 Hz and 3.2 kHz, and the mid-band edges were within the 10 % tolerance. Volume controls ranged from full mute to about 1.5× gain, keeping outputs within ±10 % of 100 mVrms. Ripple stayed under 15 mVrms, and the power amp delivered 545 mW—well above the 400 mW target.

Measured results differed only slightly from ideal values: mid-band cutoffs were 355 Hz and 3.55 kHz instead of 320 Hz and 3.2 kHz; the 1 kHz max output was 93 mVrms rather than 100 mVrms; and ripple was 13.6 mVrms instead of 15 mVrms.

These small discrepancies arose from real-world factors: resistor and capacitor tolerances (±5–10 %), limited op-amp bandwidth and slew rate, breadboard parasitics, and measurement uncertainty. Despite these, the equalizer functioned reliably and demonstrated effective analog control over audio tone.

References

Abzza, and Instructables. "Tales from the Chip: LM386 Audio Amplifier." *Instructables*, Instructables, 27 Oct. 2017, www.instructables.com/Tales-From-the-Chip-LM386-Audio-Amplifier/.